

MATH 301: Homework 3

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Sec 2.5 q2

If $S \subseteq \mathbb{R}, S \neq \emptyset$, show that S is bounded IFF $\exists I$, a closed bounded interval s.t. $S \subseteq I$

Consider the forward and reverse statements to prove:

$$\begin{cases} S \text{ is a bounded set} \implies \exists I \text{ (a closed bounded interval) } \ni S \subseteq I \\ \exists I \text{ (a closed bounded interval) } \ni S \subseteq I \implies S \text{ is a bounded set} \end{cases}$$

Suppose S is a bounded set

$$\implies \exists a = \inf S \quad b = \sup S$$

$$\implies \forall s \in S, (s \leq b) \wedge (s \geq a)$$

$$\therefore S \subseteq [a, b] \equiv S \subseteq I$$

Sec 2.5 q9

If $z \leq 0$ then $z \notin K_1$. If $w > 0$ then it follows from the Archimedian Property that there exists $n_w \in \mathbb{N}$ with $w \notin (n_w, \infty) = K_{n_w}$

Sec 3.1 q5.a

Use the definition of the limit of a sequence to establish the following limits:

$$\lim_{n \rightarrow \infty} f(n) = 0, \quad f(n) := \left(\frac{n}{n^2 + 1}\right)$$

$$\lim_{n \rightarrow \infty} f(n) = 0 \iff \forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N} \ni \forall n \geq N(\epsilon), |x_n - L| < \epsilon$$

Backwards Search

Step	Reasoning / Simplification
$\left \frac{n}{n^2+1} - 0 \right < \epsilon$	Limit definition statement
$\left \frac{n}{n^2+1} \right < \epsilon$	Subtracting zero
$\frac{ n }{n^2+1} < \epsilon$	Property of absolute values
$\frac{n}{n^2+1} < \epsilon$	Since $n \in \mathbb{N}$, n is always positive
$\frac{n}{n^2+1} < \frac{n}{n^2} = \frac{1}{n}$	Bounding: $n^2 + 1 > n^2 \implies \frac{1}{n^2+1} < \frac{1}{n^2}$
$\frac{1}{n} < \epsilon$	Sufficient condition for the inequality
$n > \frac{1}{\epsilon} \quad \frac{1}{\epsilon} \in \mathbb{R}$	
$\uparrow$	By Archimedian Property

Forward Logic: As $\frac{1}{\epsilon} \in \mathbb{R}$, this guarantees $\exists N(\epsilon) \in \mathbb{N} \ni N(\epsilon) > \frac{1}{\epsilon}$. This is the requirement to say $\forall n \geq N(\epsilon), n > \frac{1}{\epsilon}$. Therefore $\frac{1}{n} < \epsilon$. As $\frac{n}{n^2+1}$ we can generalize the apply the inequality for $f(n)$. Thus we can say $|f(n) - 0| < \epsilon$. This satisfies the epsilon definition of a limit convergence of a sequence $\therefore \lim_{n \rightarrow \infty} f(n) = 0$

Sec 3.1 q5.d

Use the definition of the limit of a sequence to establish the following limits:

$$\lim_{n \rightarrow \infty} f(n) = \frac{1}{2}, \quad f(n) := \left(\frac{n^2 - 1}{2n^2 + 3} \right)$$

$$\lim_{n \rightarrow \infty} f(n) = 1/2 \iff \forall \epsilon > 0, \exists N(\epsilon) \in \mathbb{N} \ni \forall n \geq N(\epsilon), |f(n) - L| < \epsilon$$

Backwards Search

Step	Reasoning / Simplification
$\left \frac{n^2-1}{2n^2+3} - \frac{1}{2} \right < \epsilon$	Formal definition of a limit: $ f(n) - L < \epsilon$
$\left \frac{2(n^2-1)-(2n^2+3)}{2(2n^2+3)} \right < \epsilon$	Find a common denominator: $2(2n^2 + 3)$
$\left \frac{2n^2-2-2n^2-3}{4n^2+6} \right < \epsilon$	Distribute and simplify the numerator
$\left \frac{-5}{4n^2+6} \right < \epsilon$	Combine like terms
$\frac{5}{4n^2+6} < \epsilon$	Apply absolute value: $ -5 = 5$ and $4n^2 + 6 > 0$
$\frac{5}{4n^2+6} < \frac{5}{4n^2} < \epsilon$	Bounding: $4n^2 + 6 > 4n^2 \implies$ fraction is smaller
$\frac{5}{4n^2} \leq \frac{5}{4n} < \epsilon$	Since $n \geq 1, n^2 \geq n$, so $\frac{1}{n^2} \leq \frac{1}{n}$
$\frac{5}{4n} < \epsilon$	Sufficient condition to satisfy original inequality
$\frac{1}{n} < \frac{4\epsilon}{5}$	Multiply by $\frac{4}{5}$ on both sides
$n > \frac{5}{4\epsilon}$	Take the reciprocal (flips the inequality sign)

↑ Choose $N > \frac{5}{4\epsilon}$ by the Archimedean Property

Forward Logic: Let $\epsilon > 0$ be given. As $\frac{5}{4\epsilon} \in \mathbb{R}$, the Archimedean Property guarantees there exists a natural number $N(\epsilon) \in \mathbb{N}$ such that: $N(\epsilon) > \frac{5}{4\epsilon}$. This is the requirement to say that for all $n \geq N(\epsilon)$, the inequality $n > \frac{5}{4\epsilon}$ holds. By taking the reciprocal and multiplying, we obtain: $\frac{5}{4n} < \epsilon$. Since $n \geq 1$, we know $n^2 \geq n$, which implies $\frac{1}{n^2} \leq \frac{1}{n}$. Thus, we can apply the following chain of inequalities to $f(n)$: $\left| \frac{n^2-1}{2n^2+3} - \frac{1}{2} \right| = \frac{5}{4n^2+6} < \frac{5}{4n^2} \leq \frac{5}{4n}$. Because we established $\frac{5}{4n} < \epsilon$, it follows by transitivity that: $|f(n) - \frac{1}{2}| < \epsilon$. This satisfies the epsilon definition of a limit for the convergence of a sequence. $\therefore \lim_{n \rightarrow \infty} \frac{n^2-1}{2n^2+3} = \frac{1}{2}$

Sec 3.1 q8

Prove that $\lim(x_n) = 0 \iff \lim(x_n) = 0$. Give an example to show that the convergence of $(|x_n|)$ need not imply the convergence of (x_n) .

Sec 3.1 q9

Sec 3.1 q14

Sec 3.2 q2

Sec 3.2 q4

Sec 3.2 q6.b

Sec 3.2 q6.c

Sec 3.2 q9